

Facts Before Explanation: A Provenance-Tracked Pipeline for a Decade of Personal Lab Panels

Deterministic Rules Decide What Is True; the Model Only Explains

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Abstract

A person who has had blood drawn for ten years owns a decade of evidence in a form nothing can query: PDFs, portal exports, and one spreadsheet where each laboratory wrote units, reference ranges and marker names its own way, in whichever language the clinic used. Almost everyone in this position writes the same parser once, badly, and never publishes it, so the same defects are rediscovered privately each time. This paper describes a small system that turns such an archive into a provenance-tracked fact database, and argues for the decision it is built around: deterministic, testable rules decide what is mechanically true, and a language model, if added later, may only explain. Four planes separate concerns — an immutable verbatim raw layer, a canonical parsed twin rebuilt on every run, a rules plane, and a model plane deliberately shipped empty — and ambiguity is recorded as a quality-control flag on the row rather than resolved silently or dropped. The empirical content is a failure, reproducible from the published artifact alone with no private data: matching marker names to standard LOINC identifiers by substring assigns a wrong code to 13 of the 52 analytes it matches, including three distinct measurements carrying the code for haemoglobin and red blood cells *in urine* carrying the code for red blood cells in blood. The instructive part is the repair that did not work — re-ordering candidate keys longest-first turned the failing test green while changing the outcome on exactly one row of 158, leaving twelve of the thirteen untouched: a green suite certifying a class that was 92% open. The real repair was to abandon substring matching: exact normalised match only, no code at all for an unmatched analyte, assigned codes dropping from 52 to 40. Those twelve are wrong clinically, by inspection, not merely by disagreeing with the replacement rule; what the replacement rule guarantees is only that it never invents a code, which is a weaker and different claim. Two principles generalise: for medical identifiers *missing is recoverable and wrong is not*, and a fix that turns a red test green is evidence about an instance, never about a class. We separate what a reader can run from what is our word, name the studies that would turn this case report into a measurement, and report two defects found in our own artifact while writing this — an entity-resolution gap that makes our own analyte count an overcount, and a miscount in our own README of exactly the kind the paper is about. There is no evaluation here: no gold-standard comparison, no clinical validation, no users but one.

1 Introduction

Consider an ordinary situation with no research glamour in it. Someone has had routine blood work done every year or two for a decade, in more than one country and at more than one laboratory. They hold the results. They cannot ask them anything.

The obstacles are mundane and they compound. Units and reference ranges differ between laboratories for the same marker, and sometimes between draws at one laboratory; ranges are sometimes absent, sometimes prose. Names differ by language and house style — the same analyte carries an English name at one clinic, a Russian name at another, a Portuguese name at a third. Someone eventually consolidates the archive into a wide comparison spreadsheet, one row per marker and one column block per draw, because that is the shape a human can read. It is also nearly impossible to query, and it is where the data stops moving.

What happens next is the part worth publishing. The owner writes a parser: once, under time pressure, tested by looking at the output, good enough for today’s question. It is never published, being both embarrassing and entangled with private data. So the next person writes it again and rediscovers the same defects — that the hyphen in 3.9–5.5 is a range separator and not a minus sign; that the scale token in $6.2 \times 10^9/L$ belongs to the *unit* and is not a second value; that a comment sub-row has exactly the shape of an analyte row; that “C-Reactive **Protein**” will be filed under proteins by any rule naive enough to look for the word.

This paper describes a system built for that situation and published [1] with the machinery and the mapping but — by construction — none of the measurements. It is a system paper, and its contribution is not that the parsing is hard. The contribution is a boundary and the evidence for taking it seriously.

The boundary. Deterministic rules decide what is true. A language model may explain what a fact might mean; it may not create, classify, threshold or alert. This inverts the reflex of handing a capable model the messy spreadsheet directly — and large models really are strong at this kind of normalisation, which is what makes the temptation real. But a model that assigns a standard identifier assigns it plausibly whether or not it is right, leaving no residue to audit: no rule to read, no test to fail, no diff when the answer changes. That models fabricate confidently, and that medicine punishes confident fabrication asymmetrically, is documented [15, 16, 17]. Our contribution is not that observation but its consequence in a concrete architecture: a plane the model is not allowed into, and a falsifiable test of whether the line held.

Why our own failure is the evidence. We adopted this boundary not from principle but after a deterministic component — three lines of ordinary string matching — assigned the wrong medical identifier to 13 of the 52 markers it matched, silently, in a field that looks identical whether right or wrong (§5). If code we could read, test and diff failed that way and needed a human eye to catch, the case for putting the same job inside a model is weaker, not stronger.

Contributions. (i) A **four-plane architecture** for personal clinical archives, with the rule the design hangs on — map once, never discard the original — and quality-control flags as first-class data rather than dropped rows (§3). (ii) A **curation layer as data, not code**: an ordered panel classifier whose ordering is a correctness property, evidence tiers, and code seeding that stamps everything it produces unverified (§4). (iii) The **identifier-matching failure** in full and reproducibly, including the fix that went green while leaving the class open (§5). (iv) An explicit, falsifiable **model-boundary test** (§6). (v) A **reproducibility table** separating what a reader can run from what is our word, plus two defects found in our own artifact while writing this (§§7–8).

What this paper is not. An evaluation. There is no gold-standard mapping to score against, no clinical validation, no comparison with an alternative system and no user base beyond the author; §5 names the studies that would change that. Nothing here claims any output is clinically correct: the system is a doctor-preparation and self-quantification tool, not a diagnostic one.

2 Setting and requirements

The corpus that motivated the system, described only by the counts the repository itself publishes: 622 observations across 158 analyte keys and 10 draws, 2016 to 2025, consolidated by hand into one wide comparison sheet. None of it is in the repository or in this paper; what is published is the machinery and the mapping.

Four requirements followed, in the order they bit.

R1: losslessness under a wrong parser. The parser will be wrong — ours was wrong three times in the first ingestion and three more a month later, both sets published [2]. If parsing overwrites the source, every parser bug is a data-loss event and the archive degrades each time it is improved. So the canonical layer is a *twin*, never a replacement, and fixing the parser means rebuilding rather than repairing.

R2: ambiguity must survive as data. A cell that does not parse is usually dropped or coerced, and both destroy the signal that something needs a human. A row the parser did not understand is still a row, carrying a flag for what was strange about it.

R3: local by construction. Nothing uploads: standard-library Python, no dependencies, network calls or account, so the reason to trust it is not a privacy policy but the absence of code that could violate one — and at

648 lines including tests it is auditable in an afternoon.

R4: the mapping must be editable by someone who cannot read the parser. Assigning a marker to the lipid panel is clinical curation, not programming, and knowledge living in Python can only be maintained by whoever wrote the Python. So it lives in CSV and JSON: update a spreadsheet, re-run.

3 Architecture: four planes

Four planes, each with a different lifecycle. **Raw** holds every cell exactly as the laboratory wrote it, with source row and snapshot URI, and is written once and never rewritten. **Canonical** is a parsed twin of each raw row — value, comparator, unit, reference bounds, QC flag — rebuilt from raw on every run. **Rules** are deterministic and testable, and decide what is *true*. **Model** handles explanation and language, decides *nothing*, and is shipped empty; §6 gives the test for whether that boundary held.

The schema is four SQLite tables [1]. **raw_observations** holds one row per source cell, verbatim, with its row number and the URI of the frozen snapshot. **observations_canonical** holds exactly one parsed row per raw row — a correspondence the test suite asserts — with the extracted value, any comparator, raw and cleaned unit, parsed reference bounds, the laboratory’s own out-of-range asterisk, and a QC flag. **analytes** holds one row per marker: names, panel, tier, canonical unit, draft code. **meta** holds provenance — source identifier, snapshot URI, mapping and script version, and counts of what was ingested and skipped.

Quality-control flags. Seven values are defined, and their existence is the design position: *ok*, *empty*, *nonnumeric*, *derived*, *multi_value*, *embedded_unit*, *leading_zero_suspect*. *derived* marks a laboratory-calculated rather than measured value, keeping two quantities with different error behaviour distinguishable downstream; *embedded_unit* marks a unit written inside the result cell rather than its own column (an illustrative cell of that shape is 302 mg/dL; no measurement in this paper is real) — parsed, ambiguity recorded; *leading_zero_suspect* marks a value like 01.05, possibly a mangled 1.05 or a date that fell into a result column. The parser does not guess, and never raises: an unparseable cell yields a flagged row, not an exception and not a gap.

Provenance as a column, not a log line. Every canonical row traces to a specific row of a specific frozen snapshot, and the database records which mapping and script version produced it. That is what makes the archive re-derivable rather than merely stored. The idea is standard provenance practice [18, 19] and we claim no novelty for it, but it is routinely absent from personal-health tooling, where the output of a parse is typically the only artifact that survives.

4 Curation as data

Three curation artifacts ship with the system, all of them editable without touching code.

The panel classifier. An ordered list of 16 panels — glycemic, lipids, inflammation, liver, proteins, renal, electrolytes, iron, vitamins, thyroid, hormones, tumour markers, serology, autoimmune, urine and complete blood count — with 200 keywords across them, matched case-insensitively against the marker’s names, key and abbreviation joined. First hit wins, unmatched falls to *OTHER*, and the ordering is therefore a correctness property rather than a stylistic one: inflammation must precede proteins, or “C-Reactive Protein” is captured by the proteins rule. Keywords are bilingual (English and Russian) because lab exports rarely arrive in one language; adding a language means appending strings to a JSON file.

An earlier version inferred panels from document order — “the last all-capitals row seen” — and mis-filed an entire hormone and serology tail under a vitamin header [2]: a spreadsheet’s visual grouping is a plausible signal, not a contract. The running header now survives only as an approximate hint, and *meta* records that it is approximate.

Evidence tiers. Each marker carries a tier A–D saying how much the measurement is worth attending to at all — a property of the marker, not the result. Tier A is strong outcome evidence with a standardised assay; tier D is exploratory or commercial composites, not comparable between vendors, so a trend built from them measures the vendor as much as the person. Two consequences matter: tier D is a warning about comparability rather than

an insult, and a *missing* tier-A marker is itself a finding — a decade of tier-C chemistry with no ApoB in it says something the chemistry does not.

Draft codes, stamped unverified. A seed table of 38 entries maps marker names to draft LOINC codes [7] and canonical UCUM units [8]. Every code the system writes carries `loinc_verified=0`, and nothing downstream may treat that as a verified mapping: the seed is a scaffold to speed human verification, not a claim of correctness. Section 5 is why that stamp exists.

5 The failure: twelve markers wearing another marker’s code

This section is the paper’s empirical content. It is a defect, in our own deterministic code, found by a person looking at output rather than by a test.

Stage 1: a separator read as a sign. The parser’s number pattern was signed: `[-+]?\\d+(?:[. ,]\\d+)?`. On the reference range `3.9-5.5` it read the hyphen as the sign of `-5.5`, and taking minimum and maximum produced `(-5.5, 3.9)`: inverted, with a negative floor, on *every* two-sided reference range in the database. It survived because it fails silently and plausibly — nothing crashes, the row still has bounds, and the single-sided case a human eyeballs first (`<5.2`) parses correctly. The fix was a separate unsigned pattern for reference bounds, which are non-negative, so the sign had no business being in the pattern; the regression asserts that no parsed range anywhere has a negative floor — the class, not the pair.

Stage 2: a short name winning inside a longer one. Code lookup scanned the seed dictionary for a key contained in the marker’s name, in insertion order. “HDL cholesterol” is a substring of “non-HDL cholesterol” and came first, so non-HDL cholesterol was stamped with the code for HDL cholesterol. The fix sorted candidate keys longest-first, so a specific name could not lose to a shorter name inside it. The failing test went green.

Stage 3: the class was still open, and the green test hid it. Reviewing the shipped dictionary by eye afterwards, three distinct analytes were found carrying `718-7`, the code for haemoglobin: mean corpuscular haemoglobin, its concentration, and reticulocyte haemoglobin content. None is haemoglobin. In the state then shipped — after stage 2, which had repaired exactly one row — **12 of the 52 assigned codes were wrong**; the original file-order matching had 13. The twelve include free and total testosterone sharing the testosterone code, an albumin/globulin *ratio* carrying the code for albumin, and red blood cells *in urine* carrying the code for red blood cells in blood. Each is wrong clinically, on inspection of what the code denotes, not by disagreeing with some later rule.

The claim is reproducible without any private data. Because the seed table and the analyte dictionary are both published, a reader can re-run all three matching strategies over the shipped names and reproduce Table 1 in about twenty lines of standard-library Python; the procedure is stated in §7. This matters for a paper that publishes no measurements: the central empirical claim does not rest on the author’s word about a private corpus, because the defect lives entirely in the public mapping.

matching strategy	assigned	differ	status
substring, file order	52	13	original; shipped
substring, longest key first	52	12	first fix; shipped
exact normalised name only	40	0	current; unmatched get no code

Table 1: All three strategies run over the 158 published dictionary names. “Differ” counts codes that disagree with exact name matching, which is our reference here and *not* a clinical gold standard: the bottom row’s zero says the current rule never assigns a code it cannot match exactly, not that any code was validated. The thirteen and twelve, by contrast, were each confirmed wrong by inspection. The middle row is the finding: the fix that turned the test green changed **one row of 158** and removed **one defect of thirteen**.

The root cause was not the ordering. It was the matching strategy. Substring matching is unsafe for medical identifiers because clinically distinct markers are routinely named by extending each other: haemoglobin → mean corpuscular haemoglobin; testosterone → free testosterone; RBC → RBC (urine). Longest-first ordering helps only when the correct longer key is present in the table; when it is absent, the shorter key still wins and stamps a wrong code with no signal that it did.

The repair. Exact normalised-name match only, with normalisation limited to case, dash variants and whitespace — nothing that changes meaning, since meaning-changing normalisation is how wrong codes get assigned in the first place. An analyte absent from the seed table receives *no* code and waits for a human entry. Assigned codes fell from 52 to 40 and all twelve wrong ones are gone; the regression re-derives every shipped code from the seed table and fails if any row cannot be reproduced.

Two transferable principles. First: *for medical identifiers, missing is recoverable and wrong is not*. A blank field prompts someone to look it up; a confidently wrong standard code propagates into every downstream system that trusts standard identifiers and looks exactly like a correct one. A wrong reference range is visibly odd once you look; a wrong LOINC code is invisible. Marking every code unverified limits the blast radius — it never made a code right.

Second, and more uncomfortable: *a fix that turns a red test green is evidence about an instance, never about a class*. Stage 2’s fix was correct, minimal and tested, and it made the problem harder to see. Table 1 puts a number on how much harder: it changed the assignment on one row out of 158 and eliminated one defect out of thirteen, leaving 92% of the class in place behind a green suite. What found the remaining twelve was a person reading a CSV. We know of no cheap general remedy; the specific one we adopted is to make the regression re-derive the *entire* shipped artifact from its inputs and fail if any row cannot be reproduced, rather than assert on the case that failed.

What would settle this properly. The above is a case report, not a measurement. Three studies would convert it, none needing private data. (i) **Accuracy against a blinded expert:** have a clinician or laboratory informatician independently code the 158 published names, then report precision and recall per strategy against that reference with Cohen’s κ — our “wrong” column uses exact match as its own reference, which measures self-consistency, not correctness. (ii) **An independent corpus:** run the three strategies over a public terminology set the seed table was not built from, to see whether the failure rate generalises or is an artifact of one person’s dictionary. (iii) **Property-based testing:** attack range parsing with generated inputs across Unicode dash variants, decimal-comma locales, one-sided bounds and embedded scale tokens, rather than the fixed cases our 28 assertions cover. Until (i) and (ii) exist, the honest description of §5 is a reproducible defect taxonomy from a single artifact: an effect size we can count, a generality we cannot.

6 The model plane, and how to tell whether the line held

The model plane is shipped empty: nothing in the released system calls a language model. The position is not that models are useless here — explanation in accessible language is the part a person actually wants — but that the ordering matters. Facts are established first, by rules; explanation sits strictly above them.

The rules of engagement the artifact states for any model added later are constraints, not aspirations [3]: state the evidence level with every claim; never frame a substance as a solution; never invent a number, range or citation; keep measured and derived distinguishable; name risks and monitoring; consider interactions before additions. The one that carries the architecture is the last: *no alert, no threshold and no classification may originate in a language model*.

The test. Remove the model. If any fact, classification or alert changes or disappears, the boundary has been violated. This is falsifiable, cheap, and the only claim here we would defend as a design contribution rather than a description: it turns “we use the LLM responsibly” — an intention, unobservable from outside — into a property that either holds or does not.

The alerting that would sit on the rules plane is specified but not built, and we state it at that status [4]: silent by default, requiring agreement of independent gates rather than one threshold crossing, and backtested against the person’s full history before shipping — an alert rule never run against the past is a guess wearing a uniform. Nothing in it involves a language model, which is the point.

7 What a reader can verify

The repository is standard-library Python: no dependencies, no network calls, no account. Cloning it and running three commands reproduces every number in Table 2 on the synthetic example it ships.

quantity	value	how to check
system size, including tests	648 lines	<code>wc -l src/*.py tests/*.py</code>
schema tables / QC flags / panel rules / keywords / seed entries	4 / 7 / 16 / 200 / 38	<code>schema.sql, blood_ingest.py, panels.json, loinc_seed.json</code>
dictionary rows; with a draft code; tier A/B/C	158; 40; 2/5/151	parse the CSV (§8); all codes <code>loinc_verified=0</code>
codes assigned, then differing from exact, per strategy	52/52/40, then 13/12/0	<code>reproduce_misencoding.py</code>
synthetic run: draws / analytes / raw rows	3 / 21 / 61	<code>blood_ingest.py</code> summary
QC ok / nonnumeric / derived; numeric extracted	55/3/3; 58 of 61	same summary
draft codes seeded on enrichment	19 of 21	<code>blood_enrich.py</code> summary
test assertions, all passing	28	<code>tests/test_pipeline.py</code>

Table 2: Every figure is reproducible from the public repository at the commit cited in [1], and none involves any person’s measurements. The corpus counts in §2 are the exception: they describe unpublished data and are therefore our word, not something a reader can check.

Table 1 is reproduced by `tests/reproduce_misencoding.py` [5], which re-derives all three strategies from the two shipped mapping files, names the twelve wrong markers, and exits non-zero if the published figures stop reproducing. The distinction Table 2 draws is the one we would ask reviewers of similar systems to insist on: figures about the machinery are checkable by anyone, figures about a private corpus by no one, and precision in reporting the second kind does not change that.

8 Limitations, and two defects in our own artifact

Entity resolution is unsolved, and our analyte count is an overcount. Counting distinct LOINC codes in the shipped dictionary while preparing this paper, we found four codes each assigned to two rows. They are not wrong codes: they are the *same marker* under two analyte keys, because two laboratories wrote its name differently — sodium recorded once from a Russian-language sheet and once as *Natrémia*, and likewise potassium, chloride and red blood cells. Nothing merges them. The consequence is unflattering: 158 counts naming variants rather than distinct markers, so the true figure is at most 154 and possibly lower, since this method only detects duplicates among the 40 coded rows and leaves the 118 uncoded ones unchecked. Worse than the miscount, a per-marker trend built across such a pair silently splits one series into two — the failure mode a longitudinal tool exists to prevent. Deduplicating clinical entities across languages and house styles is the obvious next problem, and precisely the one the stage-3 lesson forbids solving with fuzzy matching.

Our own README miscounted the dictionary, in the way this paper is about. The README stated that the dictionary contains 161 analytes. Parsing the file yields 158 rows. The difference is not a typo: three analyte keys contain embedded newlines, inherited from spreadsheet cells where a laboratory wrapped a long name across two lines, so counting lines gives 161 and counting records gives 158. It is a small, checkable instance of this paper’s own thesis — a number produced by the convenient tool rather than the correct one, plausible enough that nobody looked — surviving in an artifact whose entire subject is that class of error. It is corrected in the commit accompanying this preprint, with the cause noted in place rather than quietly overwritten, as is a stale clone URL left behind by a repository rename.

Other limitations, stated plainly. No LOINC code is verified; all 40 are drafts. Tier assignment is coarse — 151 of 158 markers sit at default tier C, so the field informs mainly at its extremes. The panel classifier is first-hit-wins over hand-written keywords, so correctness depends on rule order: a hazard documented rather than removed. A single bare reference bound is stored as an upper bound, which is a guess, and the raw string is kept precisely because it is one. There is no gold-standard mapping, no clinical validation, no second laboratory’s export in the corpus and no user but the author. The rule engine, explanation layer and alerting are designed and unbuilt

[4]; there are no wearable connectors, no OCR and no FHIR export, though the schema was shaped with FHIR Observation [10] in mind. The system is not a physician, offers no diagnosis or treatment guidance, and computes no single “health score”: compressing a body to one digit hides more than it shows.

9 Related work

Personal informatics. The stage model identifies collection, integration, reflection and action, and finds most tools failing at integration with barriers cascading forward [13, 14] — exactly the gap here, since collection is solved by the healthcare system itself. That self-tracking is lived and interpretive rather than purely analytic [14] argues for an explanation layer, and, we would add, for keeping it above the facts rather than inside them.

Clinical normalisation, and models in medicine. LOINC [7] and UCUM [8] are the standards seeded against; OMOP CDM [9] and FHIR [10] are what this would export to. Mapping local laboratory codes to LOINC is well studied at institutional scale, where it is labour-intensive and requires expert review [11], and cross-institutional audits find inconsistent and incorrect use even after mapping [12]. We offer no better method — ours is deliberately the least clever available — only a report of what the naive one does at personal scale, where there is no informatics department and the person doing the mapping is the person the data is about. Language models are attractive for precisely this task, performing strongly on medical benchmarks [17]; the counterweight is that fabrication is a general property of the technology [15] and that medical hallucination benchmarks show reasoning failures persisting where confident wrongness costs most [16]. Our position is narrower than a general caution and is architectural (§6). Finally, the raw/canonical split with a rebuildable derived layer is standard provenance practice [18, 19], novel in nothing but its application: here the source is irreplaceable and the parser amateur by necessity, so losslessness binds harder than where the input can simply be re-fetched.

10 Conclusion

The system is small and deliberately unclever: 648 lines, four tables, no dependencies, a curation layer that is mostly CSV, and a model plane with nothing in it. Its argument is that for a personal clinical archive this ordering is the right one — facts first, from rules a person can read and a test can re-derive; explanation second, structurally unable to change what the facts are. The evidence is our own failure rather than a benchmark: three lines of string matching assigned the wrong standard identifier to twelve markers, the fix that turned the test green repaired one row of 158, and a human eye found the rest. Those defects were findable only because there was a rule to read and an output to re-derive. Had a model assigned the codes, the wrong twelve would have looked exactly the same, been differently wrong on the next run, and left nothing to inspect — which is the case for a boundary you can falsify by deleting the model.

Code, data and contact

MIT-licensed at <https://github.com/tonydz/blood-panel-pipeline> [1]: standard-library Python, no dependencies, no network calls, no measurements — machinery and mapping only, and that will not change. The failure log of §5 is `docs/DEVLOG.md` [2]; Table 1 is re-derived by `tests/reproduce_misencoding.py` [5] in a second with no arguments; the rules of §6 are `docs/METHOD.md` [3]; what is designed and unbuilt is `ROADMAP.md` [4].

The most useful contribution a reader can make is a lab export whose layout breaks the parser, or a wrong seed entry with the correct code and a citation. Corrections are valued above stars and a failed reproduction above both: open an issue. Author: Anton Dzyatkovskiy, dzyatkovskiy.a2@gmail.com · ORCID 0000-0001-7408-3054 · <https://github.com/tonydz> · <https://tonydz.github.io>. Companion preprints from the same operation: [6].

Medical disclaimer. This is not medical advice, and the system described is not a diagnostic device. It organises evidence to bring to a clinician; it does not substitute for one. No LOINC code produced by it is verified, and none should be treated as clinically authoritative.

Authorship and responsibility. This paper and the accompanying repository are drafted by an AI agent operating as the author’s synthetic co-founder, under the author’s editorial control; the author is the responsible human for

every claim made here. Commits carry provenance trailers naming tool, model, machine and operator, and corrections are logged publicly.

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- [2] docs/DEVLOG.md in [1]: bugs that shipped before tests caught them.
- [3] docs/METHOD.md in [1]: order of operations and rules of engagement for any model or assistant.
- [4] ROADMAP.md in [1]: what exists and what does not.
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